

Supplementary Information

Cross-dimensional machine learning of in-plane dielectric constants in two-dimensional materials from three-dimensional van der Waals crystals

This Supplementary Information consolidates the materials cited collectively in the main manuscript. It contains the definitions and mathematical expressions of the physically reconstructed descriptors, the Pearson-correlation analysis for representative descriptors, and the abbreviation map for the feature names used in the feature-ranking and distribution analyses.

Supplementary Table S1| Physically reconstructed features and mathematical expressions

The physically reconstructed descriptors below were introduced to express the geometry and response-relevant density of layered crystals and monolayers on a common effective-thickness basis. Roman text is used for descriptor labels and non-variable subscripts, whereas mathematical variables are italicized.

Feature name	Symbol	Mathematical definition	Physical meaning
Effective thickness	t_{eff}	$t_{\text{eff}} = (z_{\text{max}} - z_{\text{min}}) + r_{\text{top}}^{\text{vdW}} + r_{\text{bot}}^{\text{vdW}}$	Intrinsic layer thickness after removing the artificial vacuum contribution and adding the Bondi van der Waals radii of the two boundary atoms.
In-plane area	$A_{2\text{D}}$	$A_{2\text{D}} = ab \sin \gamma$	Projected area of the two-dimensional unit cell in the xy plane.
Effective volume	V_{eff}	$V_{\text{eff}} = A_{2\text{D}} t_{\text{eff}}$	Vacuum-independent three-dimensional volume intrinsically occupied by the material.
Effective density	ρ_{eff}	$\rho_{\text{eff}} = \frac{M_{\text{cell}}}{V_{\text{eff}}} \times 1.660539$	Mass density distributed within the effective volume, expressed in $\text{g} \cdot \text{cm}^{-3}$.
Per-atom effective volume	v_{eff}	$v_{\text{eff}} = V_{\text{eff}} / N_{\text{atoms}}$	Average effective space assigned to each atom.
Structural dimensionality ratio	SDR	$\text{SDR} = \frac{t_{\text{eff}}}{a+b}$	Geometric anisotropy measure for a two-dimensional layered

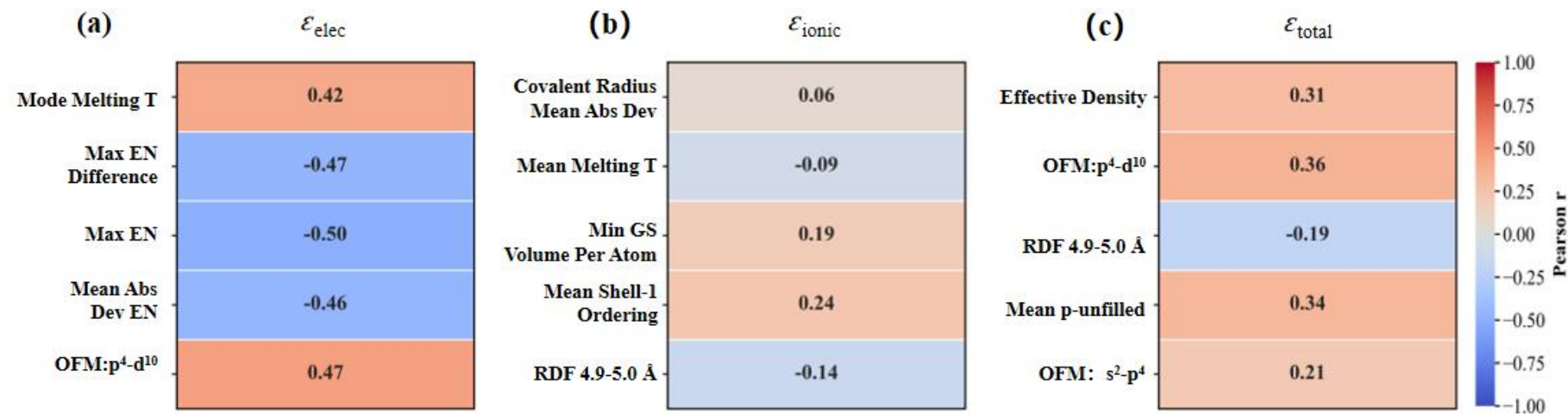
Feature name	Symbol	Mathematical definition	Physical meaning
			structure.
Area-thickness ratio	AI	$AI = \frac{A_{2D}}{t_{\text{eff}}}$	Relative proportion between in-plane extension and out-of-plane confinement.
van der Waals packing fraction	ϕ_{EPF}	$\phi_{\text{EPF}} = \frac{\sum_{i=1}^N \frac{4}{3} \pi (r_i^{\text{vdW}})^3}{V_{\text{eff}}}$	Fraction of the effective volume occupied by atomic van der Waals hard spheres.
Polarizable volume factor	α_{EAP}	$\alpha_{\text{EAP}} = \frac{\sum_{i=1}^N (r_i^{\text{vdW}})^3}{V_{\text{eff}}}$	Structural weighting factor positively correlated with macroscopic electronic polarizability.
Interfacial bonding strength	IBS	$IBS = \frac{\rho_{\text{eff}}}{t_{\text{eff}}}$	Mass packing density per unit thickness, associated with intralayer chemical-bond strength.
Effective dielectric volume	V_{EDV}	$V_{\text{EDV}} = V_{\text{eff}} f_{\text{NM}}$	Polarization-response volume dominated by nonmetal atoms, where f_{NM} is the nonmetal atomic fraction.
Maximum electronegativity difference	$\Delta\chi_{\text{max}}$	$\Delta\chi_{\text{max}} = \max(\{\chi_i\}) - \min(\{\chi_i\})$	Range of Pauling electronegativities among elements in the unit cell, indicating the tendency toward ionicity.
Average bond length	$\langle d \rangle$	$\langle d \rangle = \frac{1}{N_b} \sum_{j=1}^{N_b} d_j$	Mean nearest-neighbor bond length determined using the CrystalNN criterion.

Feature name	Symbol	Mathematical definition	Physical meaning
Bond-length standard deviation	σ_d	$\sigma_d = \sqrt{\frac{1}{N_b - 1} \sum_{j=1}^{N_b} (d_j - \langle d \rangle)^2}$	Structural disorder of the local coordination environment.
Centrosymmetry	S_{centro}	$S_{\text{centro}} \in \{0, 1\}$	Whether the space group contains an inversion center; 1 denotes yes and 0 denotes no.
Asymmetric electronegativity index	J_χ	$J_\chi = \chi_{\text{top}} - \chi_{\text{bot}} (1 - S_{\text{centro}})$	Electronegativity difference between top and bottom surface elements, screened by symmetry.
Asymmetric radius index	J_r	$J_r = \frac{ r_{\text{top}}^{\text{vdW}} - r_{\text{bot}}^{\text{vdW}} }{\max(r_{\text{top}}^{\text{vdW}}, r_{\text{bot}}^{\text{vdW}})} (1 - S_{\text{centro}})$	Relative van der Waals radius difference between top and bottom surfaces, screened by symmetry.

Note. M_{cell} is the total atomic mass of the unit cell in atomic mass units; 1.660539 is the unit-conversion factor $\frac{1 \text{ u}/\text{\AA}^3}{1 \text{ g/cm}^3}$; r^{vdW} uses the Bondi consensus radii; and the screening factor $(1 - S_{\text{centro}})$ forces the Janus indices of centrosymmetric structures to zero.

Supplementary Fig. S1 | Pearson-correlation analysis of representative descriptors

The Pearson-correlation coefficients quantify linear associations between selected descriptors and the electronic, ionic, and total dielectric responses. Positive values indicate that the descriptor tends to increase with the corresponding dielectric contribution, whereas negative values indicate an inverse trend. The magnitude should be interpreted only as a linear correlation; nonlinear feature effects are discussed in the main text through SHAP analysis.



Supplementary Fig. S1. Grayscale rendering of Pearson-correlation coefficients for representative descriptors associated with (a) the electronic contribution, (b) the ionic contribution, and (c) the total dielectric constant.

Target response	Descriptor abbreviation	Pearson coefficient
ϵ_{elec}	Mode Melting T	0.42
ϵ_{elec}	Max EN Difference	-0.47
ϵ_{elec}	Max EN	-0.50
ϵ_{elec}	Mean Abs Dev EN	-0.46
ϵ_{elec}	OFM:p4-d10	0.47
ϵ_{ionic}	Covalent Radius Mean Abs Dev	0.06

Target response	Descriptor abbreviation	Pearson coefficient
ϵ_{ionic}	Mean Melting T	-0.09
ϵ_{ionic}	Min GS Volume Per Atom	0.19
ϵ_{ionic}	Mean Shell-1 Ordering	0.24
ϵ_{ionic}	RDF 4.9-5.0 Å	-0.14
ϵ_{total}	Effective Density	0.31
ϵ_{total}	OFM:p4-d10	0.36
ϵ_{total}	RDF 4.9-5.0 Å	-0.19
ϵ_{total}	Mean p-unfilled	0.34
ϵ_{total}	OFM:s2-p4	0.21

Supplementary Table S2 | Feature abbreviations, original feature names, and physical meanings

This table maps the abbreviated descriptors shown in the feature-ranking and distribution figures to their original feature names and physical meanings. The notation follows the naming conventions used by the feature-generation workflow.

Physical abbreviation	Original feature name	Physical meaning
Mode Melting T	MagpieData_mode_MeltingT	Mode of the melting points of constituent elements.
Max EN Difference	Max_EN_Diff	Maximum electronegativity difference between constituent atoms.
Max EN	MagpieData_maximum_Electronegativity	Maximum electronegativity among constituent elements.
Mean Abs Dev EN	MagpieData_avg_dev_Electronegativity	Mean absolute deviation of the electronegativity of constituent elements.
OFM:p4-d10	OFM_p_4_d_10	p4-d10 valence-orbital interaction or co-occurrence component in the orbital field

Physical abbreviation	Original feature name	Physical meaning
		matrix.
Covalent Radius Mean Abs Dev	MagpieData_avg_dev_CovalentRadius	Mean absolute deviation of the covalent radius of constituent elements.
Mean Melting T	MagpieData_mean_MeltingT	Mean melting point of constituent elements.
Min GS Volume Per Atom	MagpieData_minimum_GSvolume_pa	Minimum ground-state atomic volume per atom among constituent elements.
Mean Shell-1 Ordering	mean_ordering_parameter_shell_1	Mean ordering parameter of the first nearest-neighbor shell.
RDF 4.9-5.0 A	rdf__4_90000__5_00000_A	Radial distribution function intensity in the 4.9-5.0 A distance range.
Effective Density	Effective_Density	Effective density descriptor computed using the effective layer volume rather than the raw supercell volume.
Mean p-unfilled	MagpieData_mean_NpUnfilled	Mean number of unfilled p-valence states among constituent elements.
OFM:s2-p4	OFM__s_2__p_4	s2-p4 valence-orbital interaction or co-occurrence component in the orbital field matrix.